

Assignment 2 - DDL & DML

1) DDL - CREATE, ALTER & DROP

```
Last login: Thu Feb 12 18:56:27 on ttys000
[(base) vijaykrishnajathare@Vijays-MacBook-Air ~ % mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.4.8 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
[mysql> USE vijay;
Database changed
mysql> CREATE TABLE Department (
->
->     dept_id INT PRIMARY KEY,
->
->     dept_name VARCHAR(50) NOT NULL UNIQUE,
->
->     location VARCHAR(50) DEFAULT 'Head Office'
->
-> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE Employee( Emp_ID INT PRIMARY KEY, Emp_Name VARCHAR(50), Designation VARCHAR(50), Salary decimal(10,2),
-> Dept_ID INT, FOREIGN KEY (Dept_Id) references Department (Dept_Id) );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Appraisal( Apr_ID INT PRIMARY KEY, Emp_ID INT, FOREIGN KEY (Emp_Id) references Employee (Emp_id),
-> Apr_year INT CHECK (apr_year >=2020), performance_score INT CHECK (performance_score BETWEEN 1 AND 10), rating VARCHAR(20));
Query OK, 0 rows affected (0.01 sec)
```

```
[mysql> DESCRIBE Department;
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(50)	NO	UNI	NULL	
location	varchar(50)	YES		Head Office	

3 rows in set (0.01 sec)

```
[mysql> DESCRIBE Department;
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(50)	NO	UNI	NULL	
location	varchar(50)	YES		Head Office	

```
3 rows in set (0.01 sec)
```

```
[mysql> DESCRIBE Employee;
```

Field	Type	Null	Key	Default	Extra
Emp_ID	int	NO	PRI	NULL	
Emp_Name	varchar(50)	YES		NULL	
Designation	varchar(50)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
Dept_ID	int	YES	MUL	NULL	

```
5 rows in set (0.00 sec)
```

```
[mysql> DESCRIBE Appraisal;
```

Field	Type	Null	Key	Default	Extra
Apr_ID	int	NO	PRI	NULL	
Emp_ID	int	YES	MUL	NULL	
Apr_year	int	YES		NULL	
performance_score	int	YES		NULL	
rating	varchar(20)	YES		NULL	

```
5 rows in set (0.01 sec)
```

```
mysql> ALTER TABLE Department
```

```
->
```

```
[ -> ADD contact_no VARCHAR(15);
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE Department
```

```
->
```

```
[ -> DROP contact_no;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE Department
```

```
->
```

```
[ -> MODIFY dept_name VARCHAR(100);
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```

mysql> ALTER TABLE Department
->
-> RENAME COLUMN dept_name TO dname;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME TABLE Department TO Dept;
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Dept VALUES (1, 'HR', 'Mumbai'), (2, 'IT', 'Pune'), (3, 'Finance', 'Delhi');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE Dept_copy AS SELECT * FROM Dept WHERE location = 'Mumbai';
Query OK, 1 row affected (0.01 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE Dept_Copy1 AS SELECT * FROM Dept;
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> TRUNCATE TABLE Dept_copy1;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Dept_copy1;
Empty set (0.01 sec)

mysql> SELECT * FROM Dept_copy;
+-----+-----+-----+
| dept_id | dname | location |
+-----+-----+-----+
|      1 | HR    | Mumbai   |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> TRUNCATE TABLE Dept_copy;
Query OK, 0 rows affected (0.01 sec)

mysql> DROP TABLE Dept_copy;
Query OK, 0 rows affected (0.00 sec)

mysql> DROP TABLE Dept_copy1;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_dept ON Dept(dept_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

```
[mysql> SHOW INDEX FROM Dept;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
Dept	0	PRIMARY	1	dept_id	A	3	NULL	NULL		BTREE			YES	NULL
Dept	0	dept_name	1	dname	A	3	NULL	NULL	YES	BTREE			YES	NULL
Dept	1	idx_dept	1	dept_id	A	3	NULL	NULL		BTREE			YES	NULL

```
3 rows in set (0.01 sec)
```

```
[mysql> DROP INDEX idx_dept On Dept;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
[mysql> SHOW INDEX FROM Dept;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
Dept	0	PRIMARY	1	dept_id	A	3	NULL	NULL		BTREE			YES	NULL
Dept	0	dept_name	1	dname	A	3	NULL	NULL	YES	BTREE			YES	NULL

```
2 rows in set (0.00 sec)
```

2) DML - INSERT, SELECT, UPDATE, DELETE, TRUNCATE, MERGE, AGGREGATE Functions, GROUP BY, HAVING, LIMIT

```
[mysql> INSERT INTO Employee VALUES (101, 'Vijay', 'Manager', 50000, 1);
Query OK, 1 row affected (0.01 sec)
```

```
[mysql> INSERT INTO Employee
[    -> (Emp_ID, Emp_Name, Designation, Salary, Dept_ID)
[    -> VALUES (102, 'Prachi', 'Developer', 40000, 2);
Query OK, 1 row affected (0.01 sec)
```

```
[mysql> INSERT INTO Employee
[    -> VALUES
[    -> (103, 'Omraje', 'Analyst', 45000, 3),
[    -> (104, 'Sahil', 'AIML', 45000, 2);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
[mysql> SELECT * FROM Employee;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
102	Prachi	Developer	40000.00	2
103	Omraje	Analyst	45000.00	3
104	Sahil	AIML	45000.00	2

4 rows in set (0.00 sec)

```
[mysql> SELECT Emp_Name, Salary FROM Employee;
```

Emp_Name	Salary
Vijay	50000.00
Prachi	40000.00
Omraje	45000.00
Sahil	45000.00

4 rows in set (0.00 sec)

```
[mysql> SELECT * FROM Employee
-> WHERE salary > 40000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
103	Omraje	Analyst	45000.00	3
104	Sahil	AIML	45000.00	2

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
->
```

```
[ -> WHERE Salary > 40000 AND designation = 'Manager';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1

1 row in set (0.00 sec)


```
[mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE 'P%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
102	Prachi	Developer	40000.00	2

```
1 row in set (0.00 sec)
```

```
[mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '%A%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
102	Prachi	Developer	40000.00	2
103	Omraje	Analyst	45000.00	3
104	Sahil	AIML	45000.00	2

```
4 rows in set (0.00 sec)
```

```
[mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '____';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
104	Sahil	AIML	45000.00	2

```
2 rows in set (0.00 sec)
```

```
[mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '_A%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
104	Sahil	AIML	45000.00	2

```
1 row in set (0.00 sec)
```

```
[mysql> SELECT * FROM Employee
-> WHERE salary BETWEEN 40000 AND 60000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
102	Prachi	Developer	40000.00	2
103	Omraje	Analyst	45000.00	3
104	Sahil	AIML	45000.00	2

4 rows in set (0.00 sec)

```
[mysql> SELECT * FROM Employee
-> ORDER BY salary DESC;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Manager	50000.00	1
103	Omraje	Analyst	45000.00	3
104	Sahil	AIML	45000.00	2
102	Prachi	Developer	40000.00	2

4 rows in set (0.00 sec)

```
[mysql> UPDATE Employee
-> SET Salary = 60000
-> WHERE Emp_ID = 101;
```

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
[mysql> UPDATE Employee
-> SET designation='Senior Manager',
-> salary=70000
-> WHERE Emp_ID = 101;
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
[mysql> UPDATE Employee
-> SET Salary = 50000;
Query OK, 4 rows affected (0.00 sec)
Rows matched: 4  Changed: 4  Warnings: 0
```

```
[mysql> SELECT * FROM Employee;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Senior Manager	50000.00	1
102	Prachi	Developer	50000.00	2
103	Omraje	Analyst	50000.00	3
104	Sahil	AIML	50000.00	2

```
4 rows in set (0.01 sec)
```

```
[mysql> SELECT COUNT(*) FROM Employee;
```

COUNT(*)
4

```
1 row in set (0.00 sec)
```

```
[mysql> SELECT SUM(salary) FROM Employee;
```

SUM(salary)
200000.00

```
1 row in set (0.00 sec)
```

```
[mysql> SELECT AVG(salary) FROM Employee;
```

AVG(salary)
50000.000000

```
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(salary), MIN(salary) FROM Employee;
```

MAX(salary)	MIN(salary)
50000.00	50000.00

```
1 row in set (0.00 sec)
```



```
[mysql> SELECT designation, AVG(salary)
-> FROM Employee GROUP BY
-> designation HAVING
-> AVG(salary) > 30000;
```

designation	AVG(salary)
Senior Manager	50000.000000
Developer	50000.000000
Analyst	50000.000000
AIML	50000.000000

4 rows in set (0.00 sec)

```
[mysql> SELECT designation, AVG(salary)
-> FROM Employee GROUP BY designation;
```

designation	AVG(salary)
Senior Manager	50000.000000
Developer	50000.000000
Analyst	50000.000000
AIML	50000.000000

4 rows in set (0.00 sec)

```
[mysql> SELECT * FROM Employee LIMIT 3;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Senior Manager	50000.00	1
102	Prachi	Developer	50000.00	2
103	Omraje	Analyst	50000.00	3

3 rows in set (0.00 sec)

```
[mysql> DELETE FROM Employee
-> WHERE Emp_ID = 105;
Query OK, 1 row affected (0.00 sec)
```

```
[mysql> DELETE FROM NewEmployee;
Query OK, 0 rows affected (0.00 sec)
```

```
[mysql> TRUNCATE TABLE NewEmployee;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE NewEmployee (
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
->     designation VARCHAR(50),
->     salary DECIMAL(10,2)
-> );
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> INSERT INTO NewEmployee VALUES
-> (101, 'Vijay', 'Manager', 60000),
-> (105, 'Sahil', 'Analyst', 45000);
```

Query OK, 2 rows affected (0.00 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Employee (emp_id, emp_name, designation, salary)
-> SELECT emp_id, emp_name, designation, salary
-> FROM newemployee
-> ON DUPLICATE KEY UPDATE
-> salary = VALUES(salary);
```

Query OK, 3 rows affected, 1 warning (0.01 sec)

Records: 2 Duplicates: 1 Warnings: 1

```
[mysql> SELECT * FROM Employee;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	Senior Manager	60000.00	1
102	Prachi	Developer	50000.00	2
103	Omraje	Analyst	50000.00	3
104	Sahil	AIML	50000.00	2
105	Sahil	Analyst	45000.00	NULL

5 rows in set (0.00 sec)

```
[mysql> SELECT * FROM NewEmployee;
```

emp_id	emp_name	designation	salary
101	Vijay	Manager	60000.00
105	Sahil	Analyst	45000.00

2 rows in set (0.00 sec)